

NE 511: Multi-Physics of Nuclear Reactors

UNIT 3: Low Fidelity Multi-Physics Modeling

Lecture 14: POINT KINETICS AND 3D NODAL KINETICS MODELS IN THERMAL-HYDRAULIC CODES

THERMAL-HYDRAULIC CODES WITH POINT KINETICS MODELS

TRAC-PF1/MOD2 will be used as an example how the core power was determined in early system thermal-hydraulic codes

Point Reactor Kinetics in TH Analysis

TRAC-PF1/MOD2

In general, two methods are used in system thermal-hydraulic codes to calculate the power:

Method I: Input specified power

- Users specify the power as a constant value or with tables.
- The tables gives power as a function of time with linear interpolation between entries.

Method II: Solution of the point reactor kinetics equations

- PRKEs specify the time behavior of the core power level with neutronic reactivity as a driving parameter.
- Neutronic reactivity is a sum of programmed reactivity (r_{prog}) and feedback reactivity (r_{fdbk}).
- The implementation of Point Kinetics Model (PKM) in thermal-hydraulics codes is supplemented by implementation of associated models: **reactivity feedback** model and **decay heat model**.

Point Reactor Kinetics in TH Analysis

TRAC-PF1/MOD2

Neutronic reactivity

- The programmed reactivity is given as a constant value or with tables.
- Programmed reactivity defines reactivity effects not accounted for by feedback reactivity, such as:
 1. *fuel reactivity*
 2. *control rod movement*
- The code evaluates feedback reactivity based on changes in:
 1. *the core averaged fuel temperature*
 2. *coolant temperature*
 3. *coolant void fraction*
 4. *boron concentration*

Point Reactor Kinetics in TH Analysis

TRAC-PF1/MOD2

The PRKEs are a coupled set of $(I+1)$ first-order differential equations defining the total thermal fission power P and the delayed neutron precursor concentrations C_i as a function of time:

$$\frac{dP}{dt} = \frac{\rho - \beta}{\Lambda} P + \sum_{i=1}^I \lambda_i C_i + \frac{S}{\Lambda}$$

$$\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} P - \lambda_i C_i, \quad i = 1, 2, \dots, I$$

P - total thermal power (W) that results from fission occurring at time t (s);

Λ - effective prompt-neutron generation time (s);

λ_i - decay constant for the delayed-neutron precursors in group i (s^{-1});

C_i - Power of the delayed delayed-neutron precursors concentration in group i (W);

S - thermal power (W) from the external source of neutrons in the reactor cores that are producing fission

The reactor multiplication factor is:

$$k = 1 + \Delta k_{prog} + \Delta k_{fdbk}$$

The neutronic reactivity, (including both programmed reactivity and feedback reactivity) is:

$$\rho = \frac{k - 1}{k} = \rho_{prog} + \rho_{fdbk}$$

The *programmed reactivity* is an input value, based on the fuel reactivity and control rod movement:

$$\rho_{prog} = \frac{\Delta k_{prog}}{k}$$

The *feedback reactivity* is code evaluated:

$$\rho_{fdbk} = \frac{\Delta k_{fdbk}}{k}$$

Point Reactor Kinetics in TH Analysis

TRAC-PF1/MOD2

The total fraction of delayed neutrons is given by $\beta = \sum_{i=1}^I \beta_i$,

where β_i is the fraction of delayed neutrons in group i and I is the number of delayed neutron groups.

Once power, P , is computed, it is used to solve the decay heat equations:

$$\frac{dH_j}{dt} = -\lambda_j^H H_j + E_j P, \quad j = 1, 2, \dots, J$$

H_j - Energy of the decay heat precursor concentration in group j (W-s);

λ_j^H - decay constant for decay heat group j (s⁻¹);

E_j - effective energy fraction of decay heat group j ;

J - number of decay heat groups

Point Reactor Kinetics in TH Analysis

TRAC-PF1/MOD2

Solution steps:

- (1) Each j^{th} equation for the decay heat group concentration H_j is solved.
- (2) Total thermal power generated in the reactor fuel at time t is calculated taking into account:

1. *prompt fission*
2. *precursor decay*
3. *delayed fission*
4. *external source*

$$P_{eff} = P + \sum_{j=1}^J \lambda_j^H H_j$$

- (3) The following are required:

1. *number of delayed neutron groups, I*
2. *delayed neutron parameters, λ_i and β_i*
3. *number of decay heat groups, J*
4. *decay heat parameters, and E_j*
5. *total thermal power history ($P(-t)$ for $t \leq 0$) and the initial delayed neutron precursor concentrations $C_i(0)$ or decay heat concentrations $H_j(0)$*

Decay Heat Groups

TRAC-PF1/MOD2

Option 1: TRAC-PF1/MOD2 internally sets J to 69 and defines λ_j^H and ED_j

The decay heat parameters are obtained from the 1979 ANS decay heat standard:

- The fraction of fission power due to each of the three isotopes, ^{235}U , ^{239}Pu , and ^{238}U , must be provided
- To convert the MeVs of decay energy per fission (ED_j) to a fraction of the total fission energy (E_j), the MeVs per fission (Q_k) for each of the three isotopes, ^{235}U , ^{239}Pu , and ^{238}U , must be provided
- ED_j is converted to E_j using $E_j = ED_j / (\lambda_j^H Q_k)$

Option 2: If the total thermal power history is not an input, TRAC-PF1/MOD2 assumes that an initial steady-state condition exists to initialize $H_j(0)$ internally:

The code sets the time derivative of H_j to zero and calculates the initial values of the H_j as:

$$H_j(0) = \frac{P(0) E_j}{\lambda_j^H}, \quad j = 1, 2, \dots, J$$

here $P(0)$ is the initial power specified through input

Prompt-Fission Power History

TRAC-PF1/MOD2

If the total thermal power $P(-t)$ history is an input, TRAC-PF1/MOD2 evaluates $C_i(0)$ and $H_j(0)$ from

$$\frac{dC_i}{dt} = -\lambda_i C_i + \frac{\beta_i P}{\Lambda}, \quad i = 1, 2, \dots, I \quad \text{and} \quad \frac{dH_j}{dt} = -\lambda_j^H H_j + E_j P, \quad j = 1, 2, \dots, J$$

The $P(-t)$ consists of tabular data (t_l, P_l) for $l=1, 2, 3, \dots, L$, where $-t_{l+1} \leq -t_l \leq 0$

The code assumes that the total thermal power varies linearly between data pairs in the power history table:

$$P(t) = a_l + b_l t \quad \text{for} \quad -t_{l+1} \leq t \leq -t_l$$

Substituting and integrating *from* $-t_l$ to $t=0$, while assuming $C_i(-t_l)=0$, gives:

$$C_i(0) = \frac{\beta_i}{\lambda_i \Lambda} \sum_{l=1}^L \left\{ \begin{aligned} &\left(a_l - \frac{b_l}{\lambda_i} \right) (1 - \exp[-\lambda_i(t_{l+1} - t_l)]) \\ &+ b_l (t_{l+1} \exp[-\lambda_i(t_{l+1} - t_l)] - t_l) \\ &- \exp[-\lambda_i t_l] \end{aligned} \right\}$$

$$H_j(0) = \frac{E_j}{\lambda_j^H \Lambda} \left\{ \begin{aligned} &\left(a_l - \frac{b_l}{\lambda_j^H} \right) (1 - \exp[-\lambda_j^H(t_2 - t_1)]) \\ &+ b_l (t_2 \exp[-\lambda_j^H(t_2 - t_1)] - t_1) \\ &- \exp[-\lambda_j^H t_1] \end{aligned} \right\}$$

Reactivity Feedback

TRAC-PF1/MOD2

In most TH codes, the reactivity feedback model is based in the assumption that the neutron multiplication is affected only by changes in:

1. *core averaged fuel temperature T_f*
2. *core averaged coolant temperature T_c*
3. *core averaged coolant vapor fraction a*
4. *boron concentration B_m (or ratio of boron mass to liquid-coolant mass, B_r)*

We can define a reactivity coefficient for each reactivity feedback independent parameter x ($T_f, T_c, a, B_m (B_r)$) by choosing one of the following reactivity coefficient forms:

Form Number	Reactivity-Coefficient Form
0	$\frac{\partial k}{\partial x}$
1	$\frac{1}{k} \frac{\partial k}{\partial x} \cong \frac{\partial R}{\partial x}$
2	$x \frac{\partial k}{\partial x}$
3	$\frac{x}{k} \frac{\partial k}{\partial x} \cong x \frac{\partial R}{\partial x}$

Reactivity Feedback

TRAC-PF1/MOD2

Volume-weighting factors approximate the product of the local adjoint flux, the neutron flux, and the volume.

TRAC-PF1/MOD2 approximates this weighting factor by the product of:

(1) material mass, **(2)** fission power density distribution raised to a user specified power, and **(3)** mesh cell volume

The core averaging of a reactivity feedback parameter x is:

$$\bar{x} = \frac{\sum_i x_i \rho_i V_i P_i^{exp}}{\sum_i \rho_i V_i P_i^{exp}}$$

If x is T_f or T_c , ρ_i is the density of the fuel or coolant, respectively, in cell i

If x is a or the boron concentration (B_m or B_r), then ρ_i is set to 1.0 for all i

Approximation: $\left(\frac{\text{material}}{\text{mass}} \right) \times \left(\frac{\text{fission power}}{\text{density distribution}} \right) = \left(\frac{\text{user specified}}{\text{power}} \right) \times (\text{volume})$

→ weighting factor can be applied to the core averaged reactivity feedback parameters.

Any of the three spatial shapes (fuel rod radial, core radial, and core axial) in the fission power density distribution can be replaced with a different shape to weight the spatial averaging of reactivity feedback parameters over the core (these spatial shape replacements are specified through input).

Reactivity Feedback

TRAC-PF1/MOD2

Each reactivity coefficient is defined through a table of reactivity coefficient values that are dependent on all four reactivity feedback parameters.

- One value for a parameter - reactivity coefficient having no dependence on that reactivity feedback parameter.
- Two or more values for a parameter - reactivity coefficient depends on one (or more) reactivity feedback parameters.
- Linear interpolation is used to evaluate the reactivity coefficient between the table entries.
- Multidimensional linear surface interpolation can be performed when the reactivity coefficient is dependent on two or more of the reactivity feedback parameters.

Reactivity Feedback- Boron Dependence

TRAC-PF1/MOD2

Boron dependence of a reactivity coefficient is defined in terms of:

1. Boron concentration in the coolant channel, B_m , in units of $kg-m^{-3}$ (macroscopic density)
2. Boron mass to coolant mass ratio, B_r , in units of ppm - **p**arts **b**oron **p**er **m**illion parts liquid coolant.

- !! If the coefficient is defined as a derivative of $B_r \rightarrow$ the boron reactivity coefficient is based only on the change in the **amount of boron, without** changing T_c and α .
- !! The boron dependence of a reactivity coefficient is especially important when the reactivity coefficient is sensitive to the shift in the neutron energy spectrum.

$\rightarrow$ *The shift in the spectrum is characterized through the proportionality to the amount of neutron capture in boron (B_m or B_r).*

Reactivity Feedback- Boron Dependence

TRAC-PF1/MOD2

The boron dependence of the reactivity coefficients is evaluated for:

1. *dissolved boric acid in the liquid coolant*
 2. *boric acid plated on core structure*
 3. *borosilicate glass in burnable-poison pins*
 4. *boron oxide in control rod cluster pins*
- The first two forms are based on boron convection into the reactor core and on plate-out of boric acid if coolant dries out.
 - The user defines the burnable poison pin and control rod cluster pin forms through input with what would be their equivalent boron concentrations in the coolant.

Reactivity Feedback- Boron Dependence

TRAC-PF1/MOD2

Whereas the physical amount of boron in the burnable poison pins is constant during the transient, its equivalent concentration for neutron capture increases with coolant temperature because of reduced spatial self-shielding of boron to neutrons in those pins

A first-order polynomial function of the core averaged coolant temperature T_c describes the equivalent burnable poison pin boron concentration

$$BPP = BPP_0 + BPP_1 \cdot T_c$$

where BPP is an equivalent burnable poison pin boron concentration; BPP_0 and BPP_1 are polynomial coefficients for BPP

Reactivity Feedback- Boron Dependence

TRAC-PF1/MOD2

The amount of control rod cluster boron in the reactor core depends on the amount of control rod insertion into the core.

Its equivalent concentration is input as a first order polynomial function of programmed reactivity ρ_{prog}

$$BCR = BCR_0 + BCR_1 \cdot \rho_{prog}$$

where BCR is a control rod cluster pin boron concentration; BCR_0 and BCR_1 polynomial coefficients for BCR

Both BPP and BCR have units of macroscopic boron density ($\text{kg}\cdot\text{m}^{-3}$) in the coolant-channel volume.

- Programmed reactivity is assumed to be proportional to the amount of effective boron mass change caused by the control rod movement in the core.

Control rod movement changes the amount of boron, but the reactivity change associated with this change is accounted for in the user defined programmed reactivity r_{prog} .

- Only the dissolved boric acid and that plated on solid structures in the reactor core are included in the evaluation of B_m or B_r for the boron reactivity coefficient derivative parameter, because the amount of boron in burnable poison pins is constant during the transient.

Reactivity Feedback

TRAC-PF1/MOD2

- After the four reactivity coefficients are evaluated by linear interpolation in their four dimensionally dependent tables, feedback reactivity is evaluated in terms of its change in the neutron multiplication over the last time step, Δt^{n-1} :

$$k^n R_{fdbk}^n = \sum_{j=1}^4 (\Delta k_{fdbk})_j \qquad \text{where} \quad x_1 = T_f \quad x_2 = T_c \quad x_3 = a \quad x_4 = B_m \text{ (or } B_r)$$

$$(\Delta k_{fdbk})_j = \frac{1}{2} \left[\left(\frac{\partial k}{\partial x_j} \right)^n + \left(\frac{\partial k}{\partial x_j} \right)^{n-1} \right] \times [x_j^n - x_j^{n-1}]$$

- The code obtains the value of $(\partial k / \partial x_i)^n$ by multiplying the reactivity coefficients defined by reactivity coefficient forms 1, 2, and 3 by (k^{n-1}) , $(1/x_i^n)$, and (k^{n-1}/x_i^n) , respectively.

Form Number	Reactivity-Coefficient Form
0	$\frac{\partial k}{\partial x}$
1	$\frac{1}{k} \frac{\partial k}{\partial x} \cong \frac{\partial R}{\partial x}$
2	$x \frac{\partial k}{\partial x}$
3	$\frac{x}{k} \frac{\partial k}{\partial x} \cong x \frac{\partial R}{\partial x}$

It is an approximation to use (k^{n-1}) rather than k^n (which is not yet known) to convert reactivity coefficient forms 1 and 3 to $\partial k / \partial x_i$

- The values of $\partial k / \partial x_i$ and x_i at time t^n (the start of the present time step) are evaluated as described above.

Their values at time t^{n-1} are those saved from their evaluation at the start of the previous time step.

- The change in x_i over the last time step (with the other x_i parameters held constant in defining the reactivity coefficient) is defined in the last factor of the equation.

Reactivity Feedback

TRAC-PF1/MOD2

- The term B_m (or B_r) includes only dissolved boric acid in the coolant and boric acid plated on the core structure.
- There is no change in the B_r equivalent boron concentration in burnable poison pins when T_f , T_c , and a are not considered to vary.
- To determine reactivity feedback during the current time step $\Delta t^n = t^{n+1} - t^n$, the end-of-time-step values of T_f , T_c , a , B_m or B_r need to be known .
- To evaluate that reactor core state requires knowing the current core state and the total energy generation rate $P_{eff}(t)$, over the time step Δt^n .
- The neutronic reactivity defined by the reactivity feedback contribution, $\rho_{fdbk}(t)$, must be known to determine the value of $P_{eff}(t)$ from the solution of PRKEs .
- The difficulty of needing to know the reactivity feedback in order to evaluate it, can be handled by assuming its Δk feedback reactivity rate is zero.
- The Δk programmed reactivity rate is assumed to be the same as in the previous time step Δt^{n-1} .

Reactivity Feedback

TRAC-PF1/MOD2

- We estimate $(k_{prog})^n$ because the independent variable needed to evaluate its tabular definition generally is a function of the reactor-core state
- A similar approximation for $(\Delta k_{fdbk}^{est})^n$ have been tested, but near steady state it caused the sign of to change each time step
- It is better to assume that $(\Delta k_{fdbk}^{est})^n$ is zero than to estimate its value based on its value in the previous time step
- Thus, estimates for the reactor multiplication constant and neutronic reactivity for the reactor state at the end of the present time step are

$$(k_{prog}^{est})^n = \Delta t^n \left\{ \frac{[k_{prog}^n - k_{prog}^{n-1}]}{\Delta t^{n-1}} \right\}$$

$$(k^{est})^n = k^n + (\Delta k_{prog}^{est})^n$$

$$(\rho^{est})^n = \frac{[(k^{est})^n - 1]}{(k^{est})^n}$$

Reactivity Feedback

TRAC-PF1/MOD2

- After each time step solution, the actual programmed reactivity is compared with its estimated value, and the discrepancy (if any) are corrected by applying the following constrain during the next time step Δt^{n+1}

$$(\Delta k^{cor})^{n+1} = [(\Delta k_{prog})^n - (\Delta k_{prog}^{est})^n] \cdot \min\left(\frac{\Delta t^n}{\Delta t^{n+1}}, 2\right)$$

If $\Delta t^{n+1} \ll \Delta t^n$, the time step reactivity correction could become very large → $(\Delta k^{cor})^{n+1}$ is limited to $2 \times (\Delta k^{cor})^n$

→
$$(k^{est})^n = k^{n-1} + (\Delta k_{prog})^n + (\Delta k^{cor})^n + (\Delta k_{fdbk})^{n-1}$$

where $k^n = k^{n-1} + (\Delta k_{prog})^n + (\Delta k_{fdbk})^n$

- !! We have assumed that $(\Delta k^{cor})^n$ and $(\Delta k_{fdbk})^{n-1}$ are small enough not to require an iterative evaluation of the present time step solution and that $(\Delta k^{cor})^n$ is large enough to be included in the next time step solution rather than be neglected.

Solution of the Point Reactor Kinetics

TRAC-PF1/MOD2

TRAC-PF1/MOD2 solves PRKEs using Kaganove method.

The derivation of the Kaganove method approximates the time dependence of P and $k_{ex} = k - 1 = \rho/(1 - \rho)$ over each integration time step by second-order polynomials and assumes that $\Lambda(t) = \ell/[1 + k_{ex}(t)]$, where ℓ is a constant.

For TRAC-PF1/MOD2, it is more appropriate to approximate:

- *time dependence of P by a second-order polynomial*
- *ρ by a first-order polynomial*
- *Λ by a constant,*

because TRAC-PF1/MOD2 linearly extrapolates its estimate of $\rho(t)$ over the fluid-dynamics time step to be evaluated and because the weak time dependence of Λ generally is unknown.

Solution of the Point Reactor Kinetics

TRAC-PF1/MOD2

TRAC-PF1/MOD2 then evaluates them for $\Delta t^n / \Delta t$ integration time steps during the Δt^n fluid-dynamics time step

We assume that $P(t) = P(0) + P_1 t + P_2 t^2$

$$\rho(t) = \rho(0) + \rho_1 t$$

and $\Lambda, \lambda_i, \beta_i, \lambda_j^H$, and E_j are constant for $0 \leq t \leq \Delta t$

$$P(0), \rho(0), \rho_1 = \left[\left(\rho^{est} \right)^n - \rho(0) / \Delta t^n \right] \quad \text{are known values}$$

Solving $\frac{dC_i}{dt} = \frac{\beta_i}{\Lambda} P - \lambda_i C_i$, $i = 1, 2, \dots, I$ for $C_i(t)$ in terms of a functional of the prompt-fission

power $P(t)$ gives:

$$C_i(t) = C_i(0) \exp[-\lambda_i t] + \frac{\beta_i}{\Lambda} \int_0^t \exp[-\lambda_i(t-\tau)] P(\tau) d\tau$$

Solution of the Point Reactor Kinetics

TRAC-PF1/MOD2

Substituting it into
$$\frac{dP}{dt} = \frac{\rho - \beta}{\Lambda} P + \sum_{i=1}^I \lambda_i C_i + \frac{S}{\Lambda}$$

and integrating the resulting equation term-by-term gives
$$P(t) = P(0) + \int_0^t \frac{\rho(\tau)P(\tau)}{\Lambda} d\tau - \sum_{i=1}^I \frac{\beta_i}{\Lambda} \int_0^t \exp[-\lambda_i(t-\tau)] P(\tau) d\tau + \sum_{i=1}^I C_i(0)[1 - \exp[-\lambda_i(t)]] + \frac{S}{\Lambda} \left[1 - \rho(0)t - \rho_1 \frac{t^2}{2} \right]$$

We can proceed further with the solution to obtain:

$$C_i(\Delta t) = C_i(0) \exp[1 - \lambda_i \Delta t I_{i_0}(\Delta t)] + \frac{\beta_i}{\Lambda} [P(0) \Delta t I_{i_0}(\Delta t) + P_1 \Delta t^2 I_{i_1}(\Delta t) + P_2 \Delta t^3 I_{i_2}(\Delta t)], \quad i = 1, 2, \dots, I$$

The code evaluates the decay-heat equations in the same way to give

$$H_j(\Delta t) = H_j(0)[1 - \lambda_j^H \Delta t I_{j_0}(\Delta t)] + E_j [P(0) \Delta t I_{j_0}(\Delta t) + P_1 \Delta t^2 I_{j_1}(\Delta t) + P_2 \Delta t^3 I_{j_2}(\Delta t)], \quad j = 1, 2, \dots, J$$

where

$$^1I_{j_m}(\Delta t) = \int_0^{\Delta t} \exp[-\lambda_j^H(\Delta t - \tau)] \tau^m d\tau$$

Point Reactor Kinetics in TRAC-PF1/MOD2

*“The core power model in TRAC-PF1/MOD2 has proved to be sufficiently general for most of the reactor safety related experiments and in reactor plants, provided that **the power shape only changes in the axial direction.**”*

*The option to specify core power directly with a tabular prescription is generally sufficient for modeling most transients of interest where **the power level behavior is known.***

*The point reactor kinetics with reactivity feedback has **demonstrated in many cases** the ability to describe the transient power response of a reactor core.*

For performing best estimate analyses of large break LOCAs, reactor kinetics has demonstrated the ability to predict the shutdown of core power by core voiding before the control rod scram is effective.”

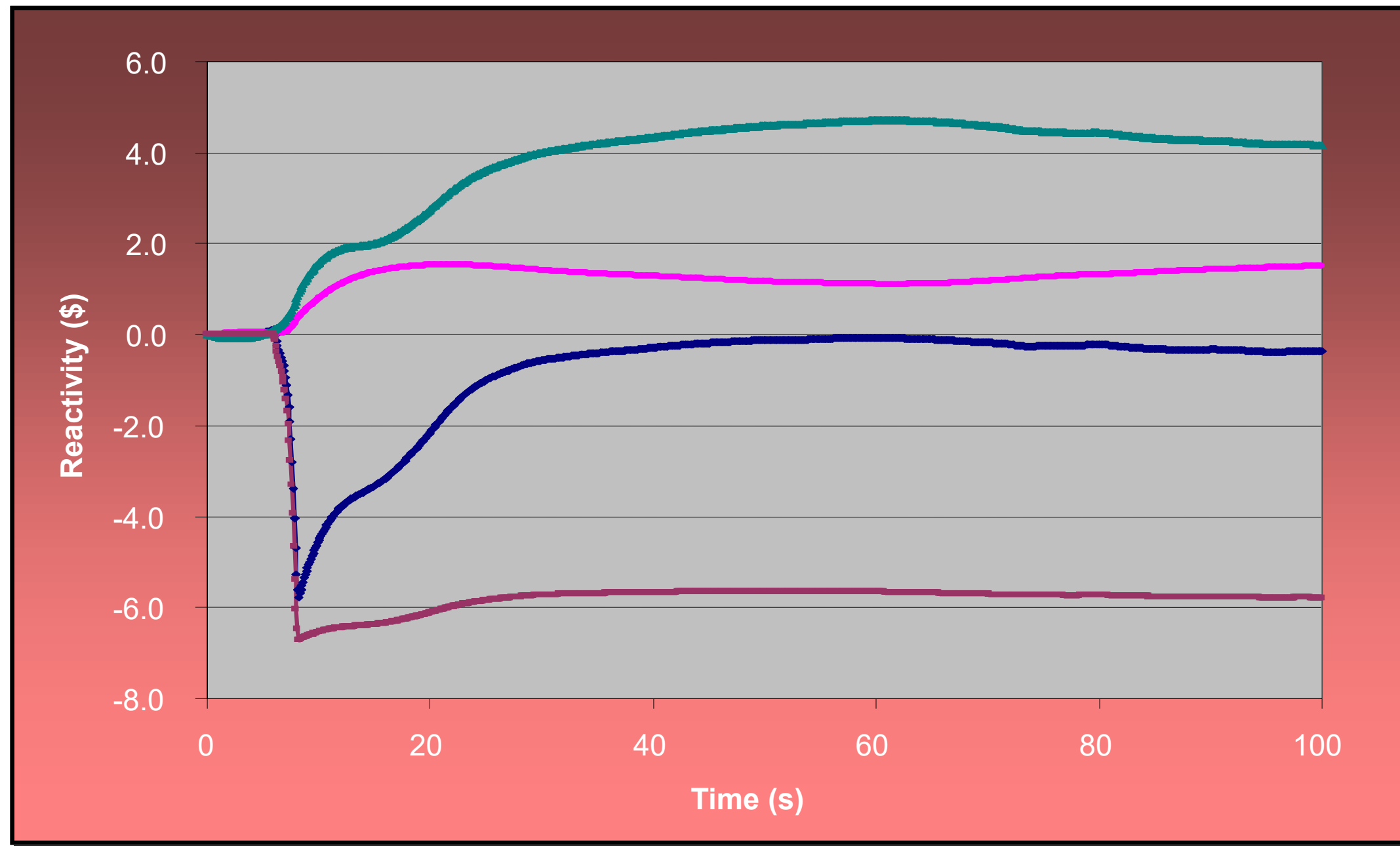
Deficiencies of Point Reactor Kinetics

- The deficiencies of the core power model are principally related to the fact that reactor kinetics is based on a point reactor assumption, and **the power distribution is defined by a superposition of fuel rod radial and core axial and $(r-\vartheta)$ plane shapes.**
- Axial shape changes in the power distribution with a known dependence can be simulated, but fuel rod radial and core $(r-\vartheta)$ plane power shape changes and power distributions that are not definable by superimposed fuel rod radial and core axial and $(r-\vartheta)$ plane shapes cannot be modeled
- Reactivity feedback is also based on the point reactor assumption with little theoretical justification for the procedure used to **core average the reactivity feedback parameters.**

Deficiencies of Point Reactor Kinetics

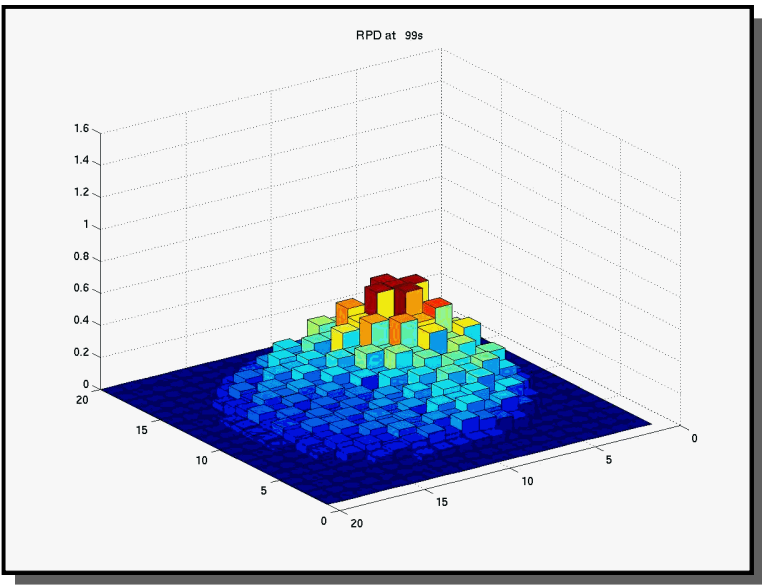
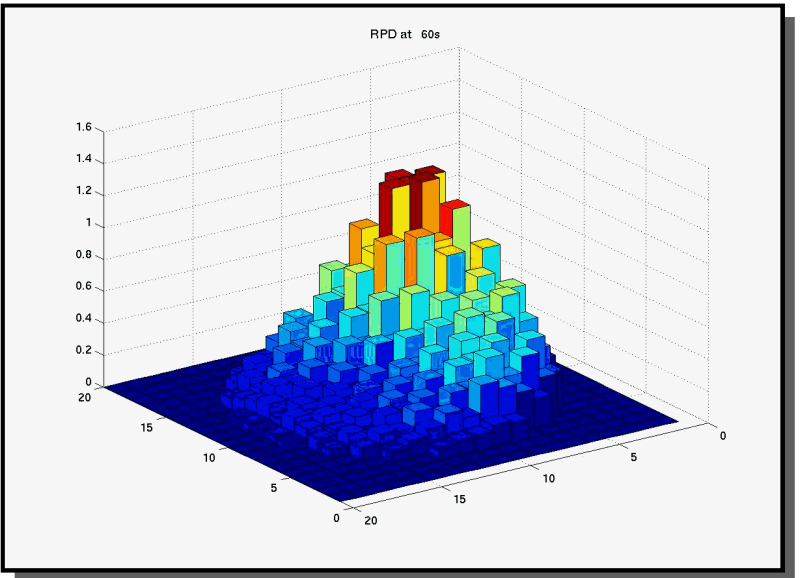
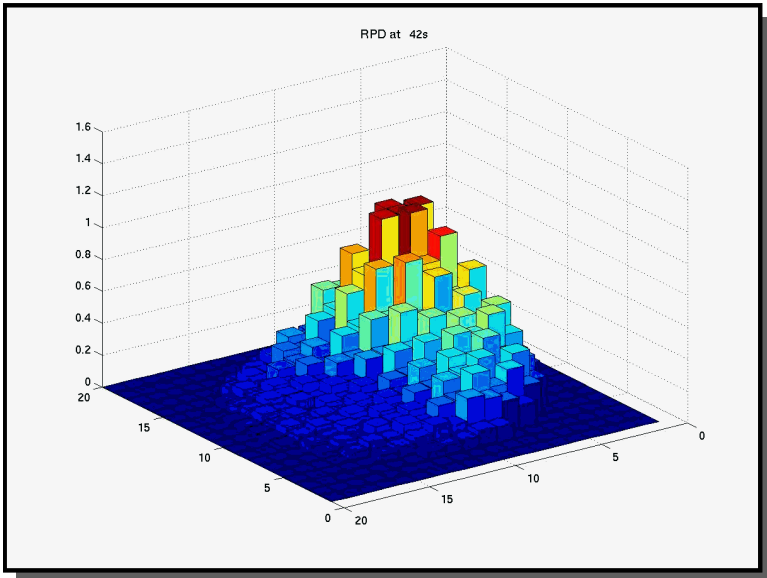
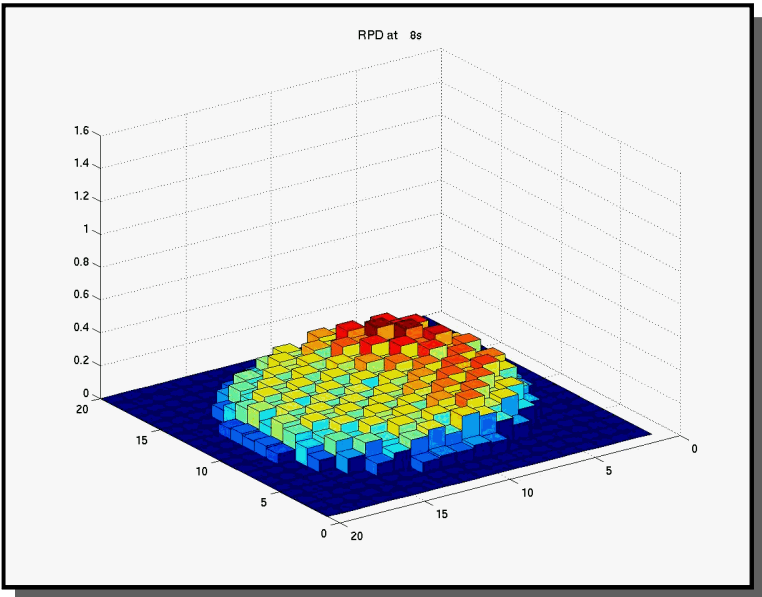
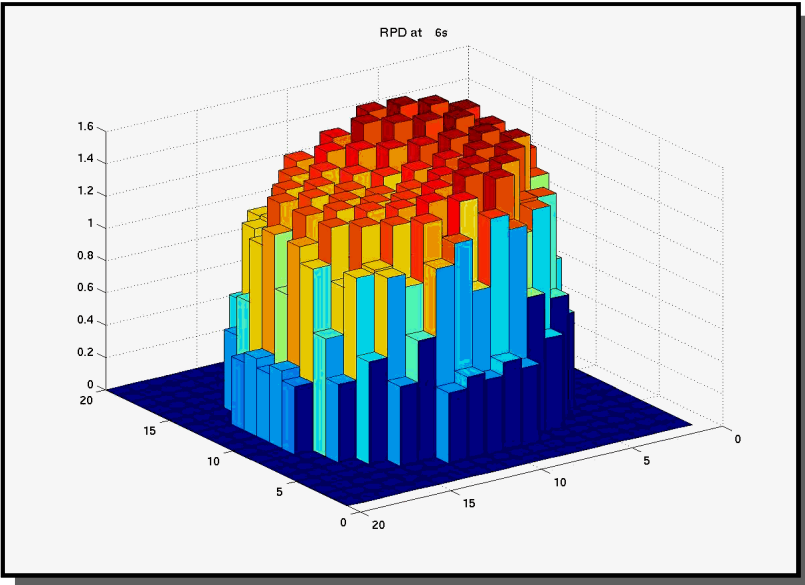
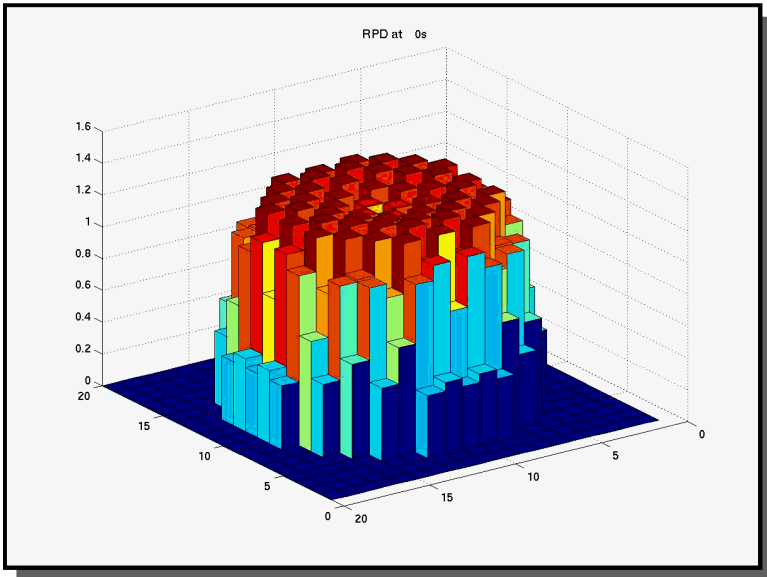
- Asymmetric and localized reactivity feedback effects and control rod movements in the reactor core cannot be modeled accurately.
- For accurate modeling of integral parameter time histories, the point reactor model needs different sets of thermal-hydraulic feedback coefficients representative of different conditions, which reactor core experiences during a transient.
- In point kinetics model, the **scram is simulated as a table** (reactivity insertion vs. time) **based on static scram evaluation at initial steady state conditions**.
 - *In 3-D kinetics, a dynamic scram is modeled, which takes into account the flux re-distribution during the scram.*
- When not affected by these modeling limitations, however, the core power model is accurate and efficient in determining the time dependent behavior of the reactor core power.

Deficiencies of Point Reactor Kinetics



3-D Kinetics Modeling

TMI-1 MSLB simulation



3D NODAL KINETICS MODELS IN THERMAL-HYDRAULIC ANALYSIS

DIRECT FINITE DIFFERENCE METHODS

Historically, the direct finite difference methods were in common use in reactor core simulations.

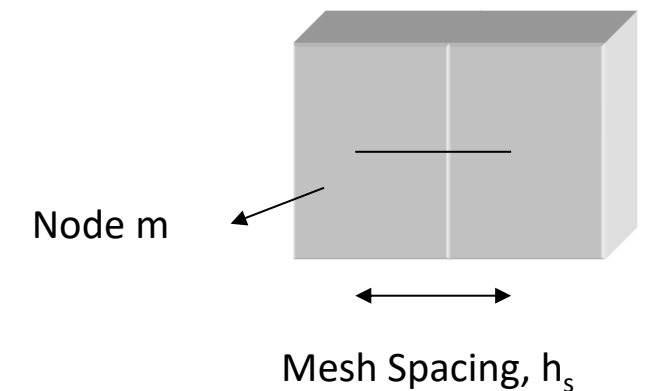
However, there is a need for more accurate methods, because:

1. The crude first order difference schemes coupled with the use of coarse meshes may become intolerably inaccurate.
2. Calculations with more refined grids are necessary for systems where noticeable heterogeneities are present.
3. Mesh refinement becomes prohibitive – for example, in a three-dimensional grid the number of unknown flux values grows rapidly with decreasing the mesh spacing.

NODAL METHODS

Nodal diffusion methods are highly competitive in LWR analysis:

- The diffusion equation simply expresses the fact that **the net production rate of neutrons in a given node must be equal the rate at which neutrons leak out of the six surfaces of the same node.**
- The nodal methods use this fact and generate a balance equation by simply integrating the diffusion equation over a node volume.
- Then, the nodal balance equation can be solved if expressions relating the surface-averaged net currents and the node-averaged scalar fluxes exists.
- The differences between various nodal methods arise from the approximations which are used to determine these nodal coupling relationships.
- The recent nodal methods are characterized by the systematic derivation of the relationship between the flux inside the node and the currents on its sides.



NODAL METHODS

Traditional nodal methods have three common features:

1. The unknowns are defined in terms of volume-averaged fluxes and/or surface averaged partial or net currents.
 2. The node fluxes and surface currents are related through auxiliary one-dimensional equations obtained by integrating the multi-dimensional diffusion equation over transverse coordinate directions.
 3. The transverse leakage term that appears in those auxiliary equations is approximated by a polynomial fit (typically quadratic) over consecutive nodes.
- Early generation nodal models assumed that the currents between nodes could be related to the difference between fluxes in neighboring nodes, that is, simple finite difference or modified coarse-mesh finite-difference approximations.
 - In the advanced nodal models, the coupling relationships can be derived by integrating the three-dimensional diffusion equation over two of the three directions of a node to obtain a transverse-integrated one-dimensional diffusion equation.

NODAL METHODS

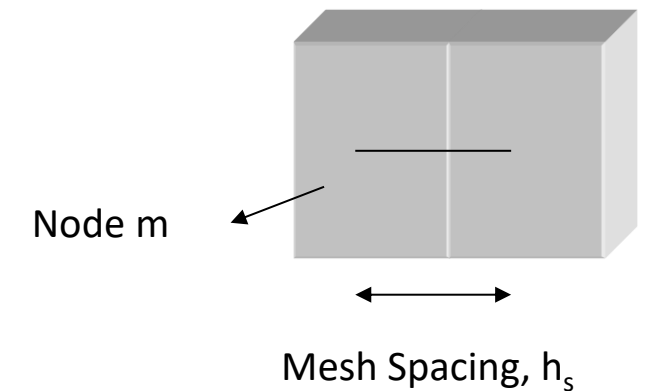
For example, in the case of the u -directed coupling relationships, one integrates over the other two directions (from $v = v^-$ to $v = v^+$ and from $w = w^-$ to $w = w^+$) to obtain

$$-\bar{D}_g^m \frac{d^2}{du^2} \phi_g^m(u) + \bar{\Sigma}_{tg}^m \phi_g^m(u) - \sum_{g'=1}^2 \left[\left(\frac{X_g}{k_{eff}} \bar{v} \bar{\Sigma}_{fg'}^m + \bar{\Sigma}_{g'g}^m \right) \phi_g^m(u) \right] = -L_g^u(u)$$

where $\phi_g^m(u) = \frac{1}{h_v h_w} \int \int \phi_g(r) dv dw$

$$\begin{aligned} L_g^m(u) &= -\frac{\bar{D}_g^m}{h_v h_w} \int \int \left[\frac{d^2}{dv^2} + \frac{d^2}{dw^2} \right] \phi_g(r) dv dw \\ &= \frac{1}{h_v h_w} \int \int \left[\frac{d}{dv} + \frac{d}{dw} \right] J_g(r) dv dw = \frac{1}{h_v h_w} \left\{ \int J_g(u, v = v^+, w) - J_g(u, v = v^-, w) dw \right. \\ &\quad \left. \int J_g(u, v, w = w^+) - J_g(u, v, w = w^-) dv \right\} \end{aligned}$$

and the mesh spacing is defined by: $h_v = v^+ - v^-$



NODAL METHODS

The nodal balance equation is

$$-\bar{D}_g^m \frac{d^2}{du^2} \phi_g^m(u) + \bar{\Sigma}_{tg}^m \phi_g^m(u) - \sum_{g'=1}^2 \left[\left(\frac{X_g}{k_{eff}} \bar{v} \bar{\Sigma}_{fg'}^m + \bar{\Sigma}_{g'g}^m \right) \bar{\phi}_g^m(u) \right] = -L_g^u(u)$$

and it is a rigorous expression for the u -dependence of the flux, which can be differentiated to obtain an expression for the net current on a u -directed surface:

$$J_g^{m,u}(u) = -\bar{D}_g^m \frac{d}{du} \phi_g^m(u)$$

Numerous methods for solving the nodal balance equation have been developed, and one efficient method introduced is **Nodal Expansion Method (NEM)**.

NODAL EXPANSION METHOD

Two approximations are required in order to derive the NEM model:

1. The transverse-integrated flux distribution within a node can be accurately represented by fifth-degree polynomials:

$$\phi_g^m(u) = \bar{\phi}_g^m + \sum_{i=1}^4 \phi_g^{i,m_u i} \quad \text{where}$$

$$\begin{aligned} u &= u & i &= 1 \\ &= 3u^2 - \frac{1}{4} & i &= 2 \\ &= u^3 - \frac{u}{4} & i &= 3 \\ &= u^4 - \frac{3u^2}{10} + \frac{1}{80} & i &= 4 \end{aligned}$$

NODAL EXPANSION METHOD

2. The u-dependence of the net leakage can be expanded in quadratic polynomials, so that the transverse-leakage shape preserves the average transverse leakage in each of three neighboring nodes in the u-directions.

In node m , the transverse-leakage shape can be expressed as:

$$L_g^m(u) = \bar{L}_g^m + \sum_{i=1}^2 l_g^{i,m_u i}$$

This approach is attractive, because as an iterative solution to the nodal balance equation is obtained:

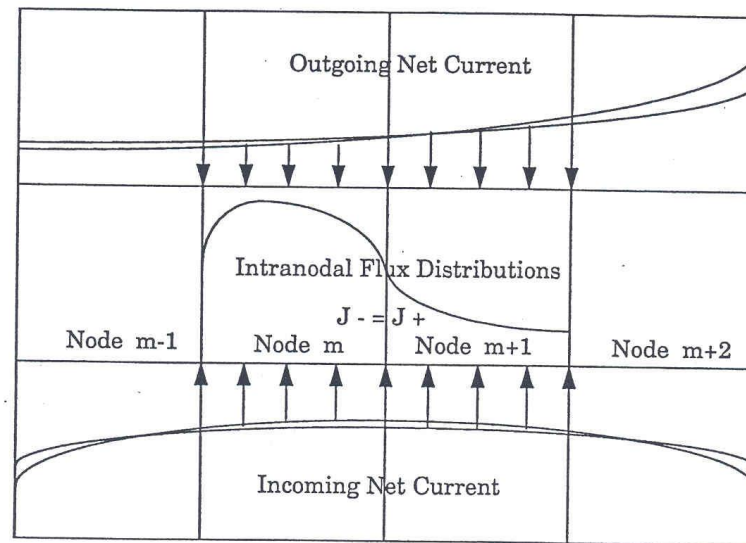
1. *the average values of the transverse leakages are known*
&
2. *the shapes of the transverse leakage can be computed directly from the node-averaged net leakages.*

NODAL EXPANSION METHOD

If the node-averaged transverse leakages are assumed to be known, the transverse integrated flux distribution within a node can be determined by substituting the last two equations into the nodal balance equation and solving the resulting polynomial form of the transverse integrated diffusion equation.

Schematically, the net current across the central interface (figure below) depends on:

1. *the shapes of the transverse-integrated fluxes*
2. *the shapes of the transverse leakages in nodes, m and $m+1$*



Two-Node Problem for the Current Across One Nodal Interface

Since the transverse-leakage shape is determined by preserving the average transverse leakage in three adjacent nodes, the expression for the interface net currents depends implicitly on the transverse leakages in four adjacent nodes in the u -direction.

NODAL EXPANSION METHOD

If the transverse leakages are assumed known, the transverse-integrated flux distribution in the two nodes can be easily determined by providing 20 constraints for the 20 unknown coefficients (2 nodes x 2 groups x 5th degree polynomials).

In the NEM model, the constraints on the flux expansion are shown below.

Node m, group g	Node m+1, group g
Node-Averaged Flux	Node-Averaged Flux
Neutron Balance	Neutron Balance
First Weighted Moment	First Weighted Moment
Second Weighted Moment	Second Weighted Moment
Continuity of Net Current	
Continuity (or Discontinuity) of Scalar Flux	

NEM NODAL SOLUTION

The NEM nodal solution is precisely the same as if the nodal equations were solved by the more conventional methods.

The iterative solution to the NEM nodal equations is obtained by use of the following procedure (non-linear iteration):

1. Initial nodal coupling coefficients are approximated as the coarse-mesh finite-difference coupling coefficients;
2. Several iterations on the nodal neutron balance equation are performed using the approximate coupling coefficients;
3. The partially converged solution is used to construct transverse-leakage shapes for each node, direction, and group;
4. Two-node problems are constructed for each nodal interface;
5. The 20 simultaneous equations are solved to obtain the expansion coefficients for the transverse integrated fluxes;
6. The flux expansion coefficients are used to compute two-group net currents across the nodal interfaces;
7. New coupling coefficients are computed for use in the iterative solution to the nodal balance equation;
8. Steps 2-7 are repeated until nodal coupling coefficients and node-averaged fluxes are converged.

NEM 3D NODAL KINETICS EQUATIONS

The TRAC-PF1/NEM coupled code is used as an example.

Two-energy groups, three-dimensional (3D) transient neutron diffusion and precursors equations for node which has constant neutron properties:

$$\frac{1}{V_1} \frac{\partial}{\partial t} \phi_1^l(r, t) - D_1^l \nabla^2 \phi_1^l(r, t) + (\Sigma_{a1}^l + \Sigma_{a2}^l - (1 - \beta) \nu \Sigma_{f1}^l) \phi_1^l(r, t) = (1 - \beta) \nu \Sigma_{f2}^l \phi_2^l(r, t) + \sum_{i=1}^I \lambda_i C_i^l(r, t)$$

$$\frac{1}{V_2} \frac{\partial}{\partial t} \phi_2^l(r, t) - D_2^l \nabla^2 \phi_2^l(r, t) + \Sigma_{a2}^l \phi_2^l(r, t) = \Sigma_{12}^l \phi_1^l(r, t)$$

$$\frac{\partial}{\partial t} C_i^l(r, t) = \beta_i (\nu \Sigma_{f1}^l \phi_1^l(r, t) + \nu \Sigma_{f2}^l \phi_2^l(r, t)) - \lambda_i C_i^l(r, t) \quad i = 1, I$$

NEM 3D NODAL KINETICS EQUATIONS

where

V_g is the energy group g neutron velocity;

$\phi_g^l(r, t)$ is the time- and space-dependent neutron flux in group g ;

D_g^l is the group g diffusion coefficient;

ν is the average number of neutrons produced by a fission event;

C_i^l is the time- and space-dependent group i delayed-neutron precursor concentration;
is the total fraction of delayed neutrons where b_i is the fraction of

$\beta = \sum_{i=1}^I \beta_i$ delayed neutrons in group i and I is the number of delayed neutron groups;

λ_i is the decay constant for delayed-neutron precursors in group i ;

Σ_{ag} is the group g absorption cross-section;

Σ_{12} is the group 1 to 2 scattering cross-section;

Σ_{fg} is group g fission cross-section

NEM 3D NODAL KINETICS EQUATIONS

First-order finite difference expression for the time derivatives of the fluxes is:

$$\frac{\partial}{\partial t} \phi_g^l(r, t) = \frac{\phi_g^l(r) - \phi_g^{l,old}(r)}{\Delta t}$$

Delayed neutron source contribution to the fast group equation is given by:

$$C_i^l(r, t^{new}) = C_i^l(r, t^{old}) e^{-\lambda_i \Delta t} + \beta_i \int_{t^{old}}^{t^{new}} (v \Sigma_{f1}^l \phi_1^l(r, t) + v \Sigma_{f2}^l \phi_2^l(r, t)) e^{-\lambda_i (t^{new} - t)} dt$$

NEM 3D NODAL KINETICS EQUATIONS

It is assumed that the fission source varies linearly between time steps:

$$\nu\Sigma_{fg}^l \phi_g^l(r, t) = \frac{\nu\Sigma_{fg}^l \phi_g^l(r) - \nu\Sigma_{fg}^{l,old} \phi_g^{l,old}(r)}{\Delta t} (t - t^{old}) + \nu\Sigma_{fg}^{l,old} \phi_g^{l,old}(r)$$

Then

$$\begin{aligned} C_i^l(r, t^{new}) = & C_i^l(r, t^{old}) e^{-\lambda_i \Delta t} + \frac{\beta_i}{\lambda_i} \left(\left(\nu\Sigma_{f1}^l \phi_1^l(r) + \nu\Sigma_{f2}^l \phi_2^l(r) \right) \left(1 - \frac{1 - e^{-\lambda_i \Delta t}}{\lambda_i \Delta t} \right) \right) \\ & + \frac{\beta_i}{\lambda_i} \left(\left(\nu\Sigma_{f1}^{l,old} \phi_1^{l,old}(r) + \nu\Sigma_{f2}^{l,old} \phi_2^{l,old}(r) \right) \left(1 - \frac{1 - e^{-\lambda_i \Delta t}}{\lambda_i \Delta t} - e^{-\lambda_i \Delta t} \right) \right) \end{aligned}$$

NEM 3D NODAL KINETICS EQUATIONS

Finally:

$$-D_g^l \nabla^2 \phi_g^l(r) + A_g^l \phi_g^l(r) = Q_g^l(r) \quad g = 1, 2$$

where

$$A_1^l = \Sigma_{a1}^l + \Sigma_{12}^l + \frac{1}{V_1 \Delta t} - (1 - \beta) \nu \Sigma_{f1}^l + \sum_{i=1}^I \beta_i \nu \Sigma_{f1}^l \left(1 - \frac{1 - e^{-\lambda_i \Delta t}}{\lambda_i \Delta t} \right)$$

$$A_2^l = \Sigma_{a2}^l + \frac{1}{V_2 \Delta t}$$

$$Q_1^l(r) = \frac{\phi_1^{l,old}(r)}{V_1 \Delta t} + (1 - \beta) \nu \Sigma_{f2}^l \phi_2^l(r) + \sum_{i=1}^I \lambda_i \left(C_i^{l,old}(r) e^{-\lambda_i \Delta t} + \frac{\beta_i}{\lambda_i} \left(\nu \Sigma_{f2}^l \phi_2^l(r) \left(1 - \frac{1 - e^{-\lambda_i \Delta t}}{\lambda_i \Delta t} \right) + \left(\nu \Sigma_{f1}^{l,old} \phi_1^{l,old}(r) + \nu \Sigma_{f2}^{l,old} \phi_2^{l,old}(r) \right) \left(\frac{1 - e^{-\lambda_i \Delta t}}{\lambda_i \Delta t} - e^{-\lambda_i \Delta t} \right) \right) \right)$$

$$Q_2^l(r) = \frac{\phi_2^{l,old}(r)}{V_2 \Delta t} + \Sigma_{12}^l \phi_1^l(r)$$

NEM 3D NODAL KINETICS EQUATIONS

- These “fixed” source equations, can be solved in either Cartesian or cylindrical geometry using the NEM spatial approximations.
- The part $Q_1^l(r)$ and $Q_2^l(r)$ of the source terms in the transient (in contrast to the steady-state) calculation are initially known (the total source magnitude is fixed and no need to calculate an eigenvalue).
- The new time flux distribution is calculated by the previous one applying inner/outer iteration strategy.
- But, to calculate the two-group flux distribution at time step one, time zero values for the average fluxes and flux moments have to be already obtained.

NEM 3D NODAL KINETICS EQUATIONS

The initial time zero fluxes are calculated by performing an initial steady-state eigenvalue calculation.

The initial precursors' concentrations are calculated from this steady-state condition by setting:

$$\frac{\partial}{\partial t} C_i^l(r, t) = \beta_i \left(\nu \Sigma_{f1}^l \phi_1^l(r, t) + \nu \Sigma_{f2}^l \phi_2^l(r, t) \right) - \lambda_i C_i^l(r, t) = 0$$

$$C_i^l(r, 0) = \frac{\beta_i}{\lambda_i} \left(\nu \Sigma_{f1}^{l,0} \phi_1^l(r, 0) + \nu \Sigma_{f2}^{l,0} \phi_2^l(r, 0) \right)$$

The old-time fluxes are calculated at the previous time step, while the previous time precursors concentrations are calculated from one and two steps previous fluxes.

TRAC-PF1/NEM COUPLING

The TRAC-PF1/NEM methodology combines:

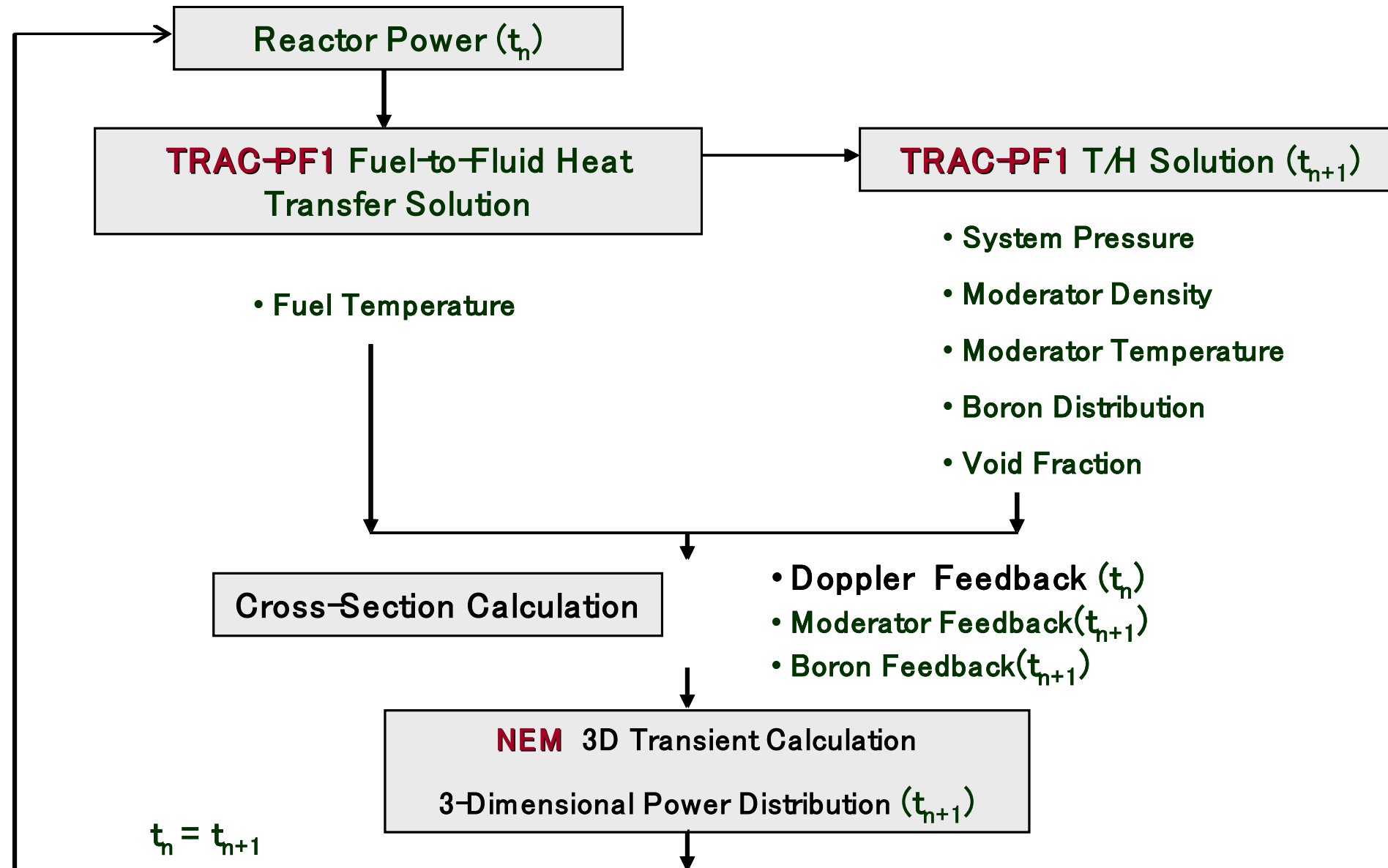
1. the 3D thermal-hydraulic models of TRAC-PF1
- &
2. a fully transient 3D neutronics description of the core.

In the TRAC-PF1/NEM coupling scheme NEM code is integrated into TRAC-PF1 as a subroutine.

The NEM input is following the TRAC-PF1 format, and the same input (TRAC-PF1) is used for the delayed neutron groups and decay heat groups.

There is a special input option which switch between Point Kinetics (TRAC-PF1) and 3D Nodal Kinetics (NEM).

TRAC-PF1/NEM COUPLING



TRAC-PF1/NEM COUPLING

Advantages of this scheme are:

1. Capability to resolve system interaction affects on the neutronics feedback of the core every time step;
2. Capability to describe local neutronics effects caused by thermal-hydraulics coupling.

Disadvantages of this scheme are:

1. Core average kinetics data (TRAC-PF1);
2. Cross-section and neutron flux data (NEM) are node wise dependent;
3. One set of kinetics, non time-dependent, core averaged parameters (delayed neutron fractions and decay constant for delayed-neutron precursors) are used in the NEM 3D equations;
4. Two-energy groups, multidimensional transient neutron diffusion equations are solved for each node but using constant neutron properties;
5. Sequential manner of coupling.

CROSS SECTION GENERATION

Two options exist:

- 1) Polynomial fitting;
- 2) Linear interpolation in four-dimensional tables.

In the polynomial fitting option cross sections are input using the polynomial-expansion procedure.

The form of the polynomial-expansion procedure is determined by the value assigned to an input variable.

According to this value four models can be used:

- 1) No thermal-hydraulic feedback is assumed:

$$\Sigma = a_1$$

- 2) Moderator Feedback:

$$\Sigma = a_1 + a_2\rho_m + a_3\rho_m^2 + a_4T_m + a_5B_m + a_6T_f^{1/2}$$

- 3) Separate Moderator and Coolant Feedback:

$$\begin{aligned}\Sigma = & a_1 + a_2\rho_m + a_3\rho_m^2 + a_4T_m + a_5B_m \\ & + a_6T_f^{1/2} + a_7\rho_c + a_8\rho_c^2 + a_9T_c + a_{10}B_c\end{aligned}$$

CROSS SECTION GENERATION

In the linear interpolation in four-dimensional tables option the cross sections are calculating by linear interpolation from multi-dimensional tables.

Up to four independent T/H parameters are used:

1. *Fuel Temperature (Doppler Feedback);*
2. *Moderator Density (T/H Feedback);*
3. *Boron Concentration;*
4. *Moderator Temperature*

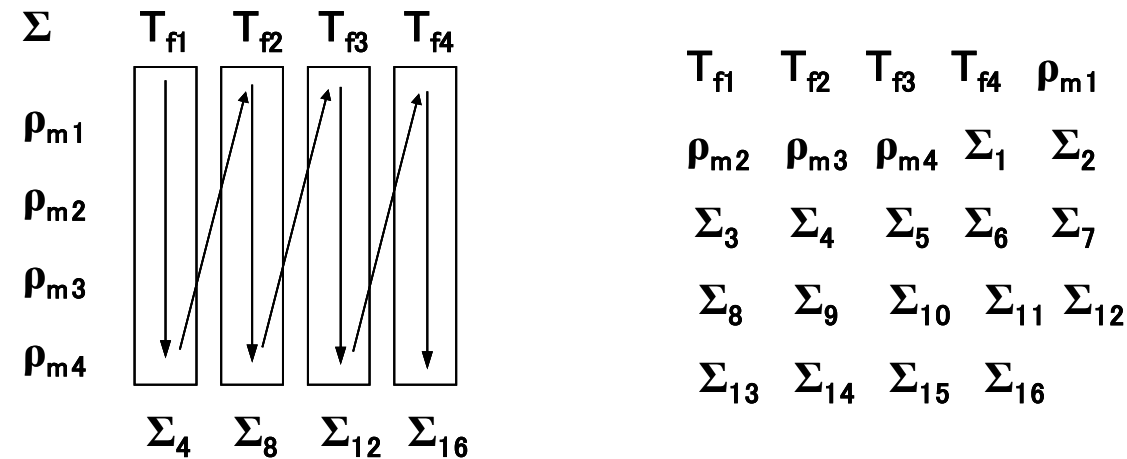
Then $\Sigma = \Sigma (T_f, \rho_m, ppm, T_m)$, where each point (T_f, ρ_m, ppm, T_m) represents one possible core state during the transient.

The expected range of the transient has to be covered by the selection of an adequate range for the independent variables.

An example of a cross-section table is shown in the next Figure.

CROSS SECTION GENERATION

4D Cross-Section Table:



- The linear interpolation in four-dimensional tables can cover a wider range of core conditions.
- There is no possibility of negative values for cross sections.
- It enhances NEM convergence rate.
- While the polynomial fitting is only accurate for small perturbations; negative values for cross sections are possible and may lead to non-converging solutions.

Next Class:

Heat Conduction and Fuel Rod Modeling in Subchannel and System TH Codes
Fuel Performance with Neutronics and Thermal-Hydraulics Feedback